

EE3621: Digital Signal Processing

Lecture 28: IIR filter design - bilinear transformation (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Bilinear Transformation algebraic mapping and frequency warping relation.
 2. **Prewarp** digital filter specifications into continuous analog frequencies.
 3. **Design** Butterworth and Chebyshev IIR digital filters using the Bilinear Transformation.
-

2. MATHEMATICAL FOUNDATIONS

2.1 Derivation via Trapezoidal Integration

Consider the first-order differential equation $\frac{dy(t)}{dt} = x(t)$. Integrating over $[(n-1)T_d, nT_d]$ using the trapezoidal rule:

$$y(nT_d) - y((n-1)T_d) = \int_{(n-1)T_d}^{nT_d} x(t)dt \approx \frac{T_d}{2}[x(nT_d) + x((n-1)T_d)]$$

In discrete notation:

$$y[n] - y[n-1] = \frac{T_d}{2}[x[n] + x[n-1]]$$

Taking the Z-Transform:

$$Y(z)(1 - z^{-1}) = \frac{T_d}{2}X(z)(1 + z^{-1}) \implies H(z) = \frac{Y(z)}{X(z)} = \frac{T_d}{2} \frac{1 + z^{-1}}{1 - z^{-1}}$$

Since the analog integrator transfer function is $H_a(s) = \frac{1}{s}$, equating $H_a(s) = H(z)$ yields:

$$s = \frac{2}{T_d} \frac{1 - z^{-1}}{1 + z^{-1}} = \frac{2}{T_d} \frac{z - 1}{z + 1}$$

2.2 Frequency Warping and Prewarping

Substitute $s = j\Omega$ and $z = e^{j\omega}$:

$$j\Omega = \frac{2}{T_d} \frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} = \frac{2}{T_d} \frac{e^{j\omega/2} - e^{-j\omega/2}}{e^{j\omega/2} + e^{-j\omega/2}} = j \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

$$\Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

- **Prewarping Formulas:**

$$\Omega_p = \frac{2}{T_d} \tan\left(\frac{\omega_p}{2}\right), \quad \Omega_s = \frac{2}{T_d} \tan\left(\frac{\omega_s}{2}\right)$$

Pictorial Representations

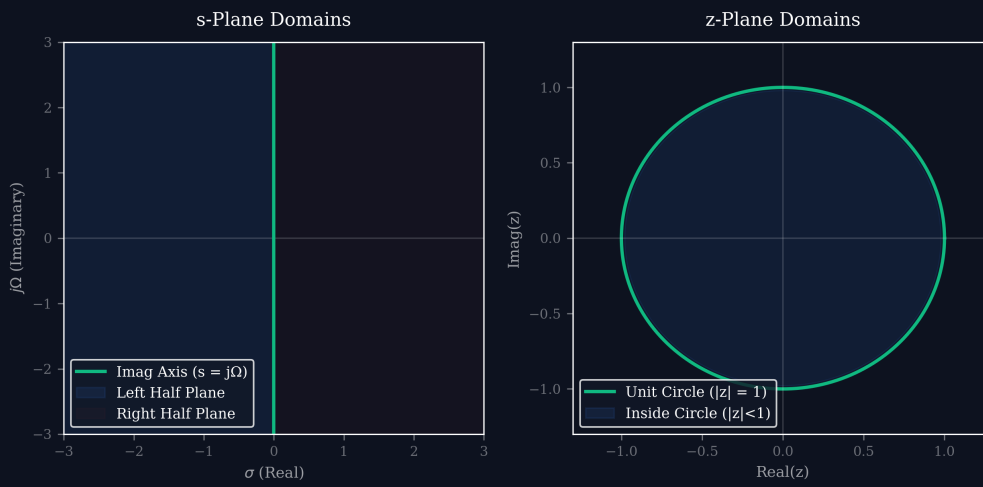


Figure 1: Bilinear S To Z

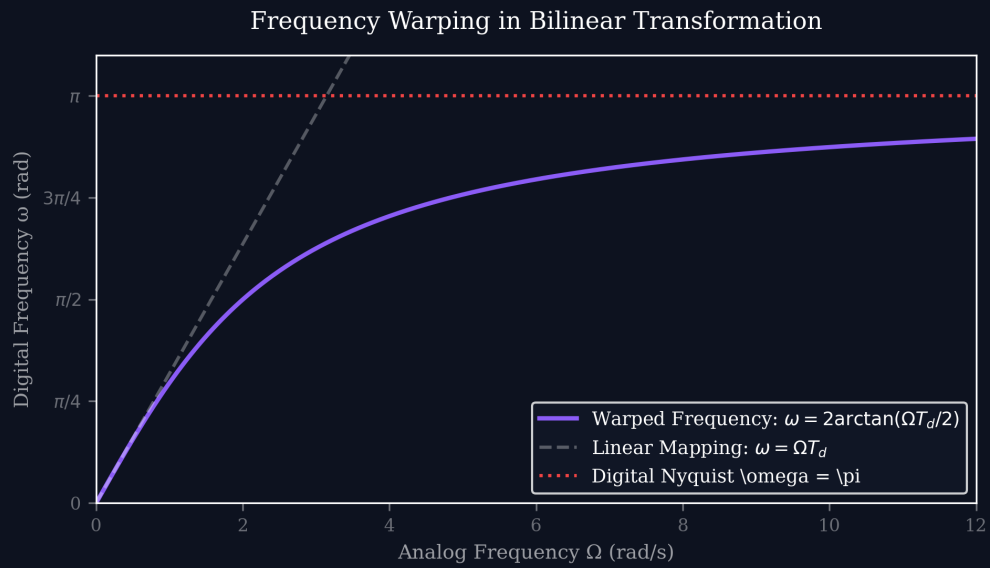


Figure 2: Bilinear Frequency Warping

3. WORKED NUMERICAL EXAMPLES

Example 28.1: 1st-Order Lowpass Filter via BLT

Problem: Design a digital lowpass filter with a 3-dB cutoff frequency $\omega_c = 0.2\pi$ rad/sample from an analog prototype $H_a(s) = \frac{\Omega_c}{s + \Omega_c}$ using the Bilinear Transformation ($T_d = 1$ s).

Solution:

1. **Prewarp the Cutoff Frequency:**

$$\Omega_c = \frac{2}{T_d} \tan\left(\frac{\omega_c}{2}\right) = \frac{2}{1} \tan(0.1\pi) = 2 \times 0.3249 = 0.6498 \text{ rad/s}$$

1. **Analog Transfer Function:**

$$H_a(s) = \frac{0.6498}{s + 0.6498}$$

1. **Apply Bilinear Transformation ($s = 2\frac{1-z^{-1}}{1+z^{-1}}$):**

$$H(z) = \frac{0.6498}{s + 0.6498} \bigg|_{s=2\frac{1-z^{-1}}{1+z^{-1}}} = \frac{0.6498}{2\frac{1-z^{-1}}{1+z^{-1}} + 0.6498} = \frac{0.6498(1+z^{-1})}{2(1-z^{-1}) + 0.6498(1+z^{-1})}$$

$$H(z) = \frac{0.6498 + 0.6498z^{-1}}{(2 + 0.6498) + (-2 + 0.6498)z^{-1}} = \frac{0.6498(1+z^{-1})}{2.6498 - 1.3502z^{-1}} = \frac{0.2452(1+z^{-1})}{1 - 0.5095z^{-1}}$$

1. **Verification:**

* At DC ($\omega = 0, z = 1$): $H(1) = \frac{0.2452(2)}{1-0.5095} = \frac{0.4904}{0.4905} \approx 1.0$ (0 dB). * At half-sampling ($\omega = \pi, z = -1$): $H(-1) = \frac{0.2452(0)}{1+0.5095} = 0$ ($-\infty$ dB).